

The number of letters in 20 students' full names is shown below.

6, 7, 8, 8, 10, 11, 12, 12, 12, 12, 13, 15, 15, 16, 18, 19, 19, 21, 24, 32

1. Determine the five-number summary for the data set.

Minimum: 6

First Quartile: 10.5

Median: 12.5

Third Quartile: 18.5

Maximum: 32

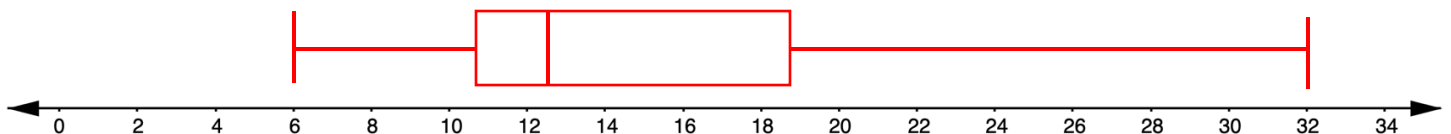
2. What is the range of the data?

$$32 - 6 = 26$$

3. What is the interquartile range of the data?

$$18.5 - 10.5 = 8$$

4. Create a box plot of the number of letters in the students' full names.



A school's data on the number of daily absences for the last 14 days are shown.

|   |   |    |   |    |   |    |   |   |   |    |    |   |    |
|---|---|----|---|----|---|----|---|---|---|----|----|---|----|
| 7 | 8 | 11 | 6 | 10 | 4 | 13 | 2 | 5 | 8 | 10 | 13 | 6 | 17 |
|---|---|----|---|----|---|----|---|---|---|----|----|---|----|

5. Determine the five-number summary for the data set.

2, 4, 5, 6, 6, 7, 8, 8, 10, 10, 11, 13, 13, 17

Minimum: 2

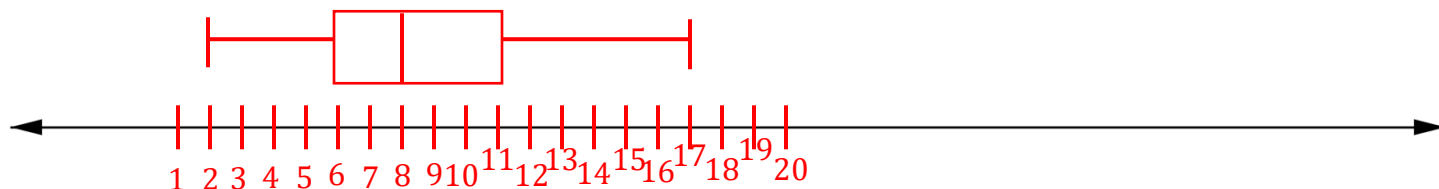
Median: 8

Maximum: 17

First Quartile: 6

Third Quartile: 11

6. Create a box plot of the daily absences.



7. Explain how you determine the scale to use for the number line of your box plot.

Use the range of data to determine an appropriate scale. All of the data values should fit on the number line.

The free-throw percentages of twelve players on a basketball team are shown.

|      |      |    |      |      |      |      |      |      |      |      |      |
|------|------|----|------|------|------|------|------|------|------|------|------|
| 71.2 | 88.7 | 50 | 81.1 | 84.3 | 78.7 | 74.5 | 82.4 | 85.1 | 91.5 | 76.5 | 87.2 |
|------|------|----|------|------|------|------|------|------|------|------|------|

8. Determine the five-number summary for the data set.

50, 71.2, 74.5, 76.5, 78.7, 81.1, 82.4, 84.3, 85.1, 87.2, 88.7, 91.5

Minimum: 50

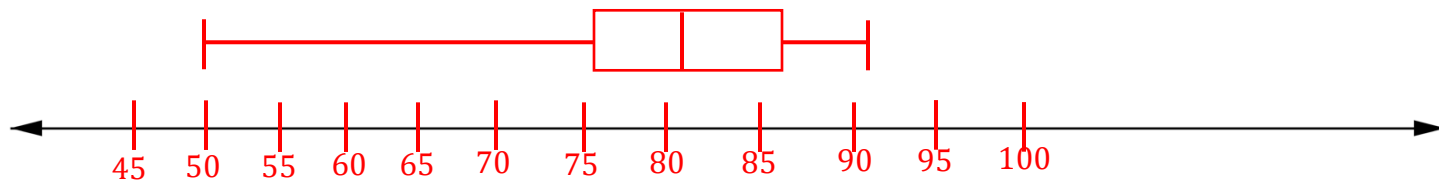
Median: 81.75

Maximum: 91.5

First Quartile: 75.5

Third Quartile: 86.15

9. Create a box plot of the team's free-throw percentages.



10. What is the interquartile range?

$$86.15 - 75.5 = 10.65$$